Streamlining Ancestry: A Lightweight Document Processing Pipeline for Historical Document Data Extraction

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by

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***Abstract*—The rise in popularity of digital genealogical libraries has brought a complexity to image databases. Records are harder to search and require indexing. We propose a pipeline to quickly extract data from scanned documents. This pipeline is transferable to many document formats and requires minimal manual labor to train the system. Our pipeline utilizes YOLOv8 for object detection and EasyOCR and TrOCR for optical character recognition. The YOLOv8 model is trained to recognize consistent machine printed fields which act as anchors for regions of interest. Our YOLOv8 model achieved a mAP50 of .995, our EasyOCR model achived a CRA of 85.4%, and our TrOCR model achieved a disappointing CRA of 17.16%. This research has concluded that cursive handwriting is difficult to recognize especially in instances where data extends into other fields. We conclude the paper by addressing the balance between data privacy and public accessibility and the responsibilities entrusted to users of this technology.**

***Index Terms*—Optical Character Recognition, Object Detection, YOLO, TrOCR, Genealogy, EasyOCR, Historical Document, Data Extraction**

I. Introduction

In a world migrating to the internet, it is important to have digital versions of historical records. Digitalization of vital records is crucial in safeguarding cultural heritage, ensuring that traditional knowledge and historical records remain accessible for future generations[1]. In recent decades there has been an influx of digitalization going on especially pertaining to genealogy[2]. Companies like FamilySearch and Ancestry boast massive databases of images of records of humans across the world. A key advancement in digital preservation is the advancement of content management. Collections can be indexed, searched, and retrieved faster than physical archives[3]. Image collections are difficult to index because programs cannot directly read scanned image documents.

Much of genealogy is moving to the digital humanities, recognition of electronic genealogy databases including production and modification of database and users digital genealogy resources are showing slow but positive inroads towards acceptance in the field. Discussion of browsing behavior, navigation, searching and metadata expands the view of digital resources beyond historians and provides clear examples of how genealogy can fit within the heterogeneous field and serve as an important source of academic scholarship[2].

Ancestry has over 40 billion records of historical documents. This data can be difficult to search at times. Historical documents (birth, marriage, death) are plentiful and take humans a long time to accurately transcribe. Manual data extraction is arduous and susceptible to human error[3]. We propose a system that can quickly and accurately turn a scanned version of a record into plain text that can be easily appended to an online database that would be easier to search. Given that there are many document formats; this system would need to be adaptable to many formats. We look to the current literature to address the problems and decide the best way to complete this task using artificial intelligence to increase efficiency and accuracy.

II. Literature Review

Handwritten text recognition provides the means to create more accurate transcriptions from existing digital images, particularly those with complex fonts or layouts. historians can begin to confidently use large datasets derived from handwritten text recognition instead of relying on in person visits to archives[4]. With recent advancements in artificial intelligence technology there are many applications in document recognition and data extraction including genealogical forms.

Our research builds on the work done by Sol et al.[5] who have laid down a framework for historical document recognition. The tools they use are different from the ones we propose; however, they have put forth an impressive framework that boasts 74% of entries considered complete and valid. Using advanced tools and a slightly different approach, we hope to achieve higher accuracy. Their complete pipeline from image to database creates a target for our further research.

This research also builds on work done by Martinez et al.[6] where they used YOLOv8 (You Only Look Once) to locate consistent, easy to find fields rather than having the program figure out what it is important. This puts more work on the human due to the labeling of the data; however, since you are selecting the same fields that are always the same size and relatively in the same space, you don't have to label a large amount of data for training. Martinez et al.[6] achieved a mean Average Precision(mAP) score of 99.42%. This process increases the accuracy of the object detection network at the cost of a few hours of human labeling data. We have decided to use YOLOv8 to continue the research due to the success of Martinez et al.[6] as well as the transfer learning approach, which allows YOLO to be applied to many different object detection tasks. The ability to select between different sizes of models from nano to extra-large allows you to decide based on the difficulty of the task or if you would like the model to run faster[7].

Another crucial part of this system is the Optical Character Recognition (OCR) systems. With the nature of documents we are working with we need models that can recognize hand-written text and models that are better at recognizing things like abbreviated words, or single characters. We have selected TrOCR (transformer based optical character recognition) for the handwritten text recognition portion. TrOCR has been found to be able to recognize text in multiple languages[8], TrOCR performs well on handwritten text as well as printed text in the case that data in the same location in the same forms may be handwritten in one and printed in another[9]. We have elected to use EasyOCR which has proved to perform well in instances of single character or non-word instances where TrOCR looks for a dictionary word EasyOCR reads the data for what it is. EasyOCR has also performed well on printed text so we will keep it in areas where we know printed text is present[10].

III. Framework

An image takes many steps through this system to reach the plain text result. You begin with an image file which is fed through YOLOv8 object detection network which returns a text file with the locations of all the labels which are focused on machine printed, relatively unchanging fields. A visualized version of this can be seen in fig. 1. The text file with the label information is fed through a script using the label data as an anchor that then mathematically calculates the offsets to extract the Region of Interest (ROI) and returns their location in a text file. A visualized version of this step can be viewed in fig. 2. The text file with the ROI is passed to the OCR part of the system where the program reads through the file sending the extracted data to be read by the system switching between TrOCR and EasyOCR. Examples of the data fed into the OCR models is available in fig 3 and 5 respectively and model predictions of this data are visible in fig. 4 and 6 respectively. The result of this process is a comma separated values (csv) file with the words that are seen on the form.

There are important human portions of this system that allow it to be malleable to the document format. The YOLOv8 network is trained on 15 bounding box labeled images. A human must spend the time correctly labeling the fields in the same order each time[11]. The object detection network only requires 15 pieces of labeled data because we are focusing on fields that are reliably similar image to image; however,

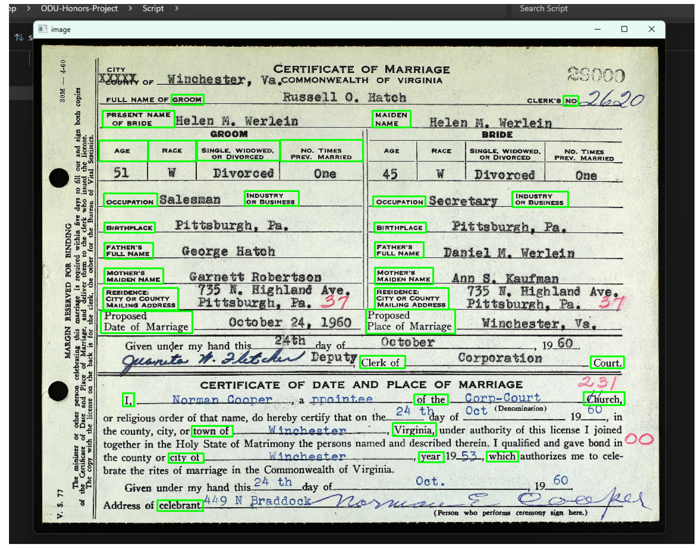


Figure 1. A piece of data after YOLOv8 has predicted the machine printed fields of interest visible with the green outline.

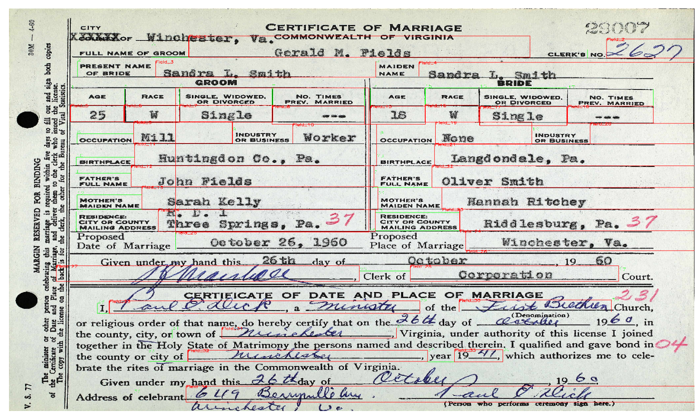


Figure 2. Image of the data after offset math occurs. Red boxes denote Regions of Interest which will be extracted and fed into an OCR model.

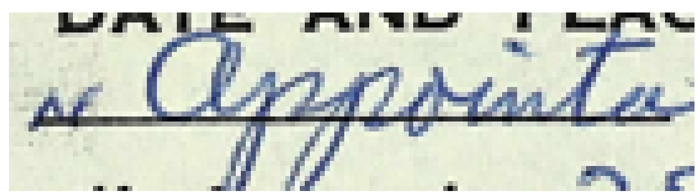


Figure 3. Spliced image of data extracted to be read by TrOCR.

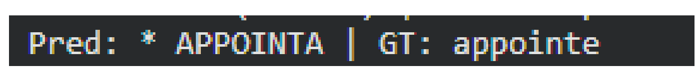


Figure 4. Pred is the characters the model predicted and GT stands for ground truth. This result is for the piece of data seen in Fig. 3

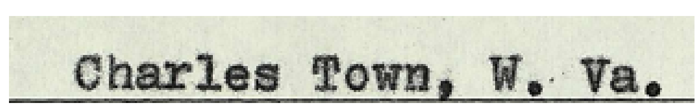


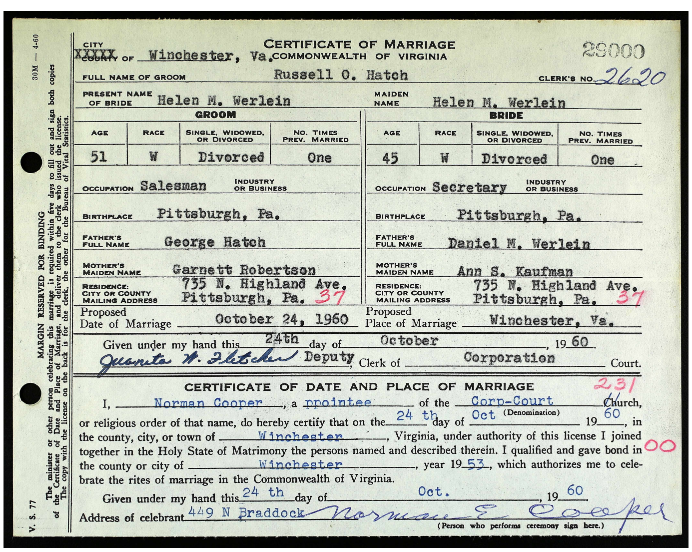
Figure 5. Spliced image of data extracted to be read by EasyOCR.

Figure 6. Predicted characters for the data extracted in Fig. 5

the use of deep learning allows us to confidently say that if there was any error with the scanning of these documents it would not mess up the system. A human is also required to configure the equations that decide the offset for the ROI[12]. In total the time that human labor is required is a fraction of the insurmountable pain of manually extracting data. All scripting was done in python and executed using Jupyter Notebook and all artificial intelligence was completed using the free T4 Graphical Processing Unit (GPU) on Google Colab[13].

IV. Dataset

The dataset we used for initial testing of our system was Virginia Marriage Certificates October 1960. We labeled 20 pieces of data in a 15:5 training and testing split. We used the default YOLOv8 augmentation except for lrflip and scale which were turned to 0.0. We recognize that 20 images is a small dataset, however, since we are transfer learning from YOLOv8m and given the predictability of our class we believe this is adequate to yield satisfactory results. For the OCR models I hand labeled ground truth for 42 pieces of data to validate EasyOCR and 27 for TrOCR[14]. Data in the case of the OCR models was spliced images from the same Virginia Marriage Certificates October 1960 data set we used for our object detection network. This dataset is also small however it is large enough to give us a feel for how the model can handle data from our specific document format.



30M — 4-60

CERTIFICATE OF MARRIAGE 29000

CITY XXXXX COUNTY OF Winchester, Va. COMMONWEALTH OF VIRGINIA

FULL NAME OF GROOM Russell O. Hatch CLERK'S NO. 2620

PRESENT NAME OF BRIDE Helen M. Werlein MAIDEN NAME Helen M. Werlein

GROOM				BRIDE			
AGE	RACE	SINGLE, WIDOWED, OR DIVORCED	NO. TIMES PREV. MARRIED	AGE	RACE	SINGLE, WIDOWED, OR DIVORCED	NO. TIMES PREV. MARRIED
51	W	Divorced	One	45	W	Divorced	One

OCCUPATION Salesman INDUSTRY OR BUSINESS — OCCUPATION Secretary INDUSTRY OR BUSINESS

BIRTHPLACE Pittsburgh, Pa. — BIRTHPLACE Pittsburgh, Pa.

FATHER'S FULL NAME George Hatch — FATHER'S FULL NAME Daniel M. Werlein

MOTHER'S MAIDEN NAME Garnett Robertson — MOTHER'S MAIDEN NAME Ann S. Kaufman

RESIDENCE: CITY OR COUNTY MAILING ADDRESS 735 N. Highland Ave. Pittsburgh, Pa. 37 — RESIDENCE: CITY OR COUNTY MAILING ADDRESS 735 N. Highland Ave. Pittsburgh, Pa. 37

Proposed Date of Marriage October 24, 1960 — Proposed Place of Marriage Winchester, Va.

Given under my hand this 24th day of October, 19 60.

Juanita W. Fletcher Deputy, Clerk of Corporation Court.

MARGIN RESERVED FOR BINDING

The minister or other person celebrating this marriage is required within five days to fill out and sign both copies of the Certificate of Date and Place of Marriage, and deliver them to the clerk who issued the license. The copy with the license on the back is for the clerk, the other for the Bureau of Vital Statistics.

CERTIFICATE OF DATE AND PLACE OF MARRIAGE 231

I, Norman Cooper, a ppointee of the Corp-Court Church, (Denomination) or religious order of that name, do hereby certify that on the 24 th day of Oct 19 60, in the county, city, or town of Winchester, Virginia, under authority of this license I joined together in the Holy State of Matrimony the persons named and described therein. I qualified and gave bond in 00 the county or city of Winchester, year 19 53, which authorizes me to celebrate the rites of marriage in the Commonwealth of Virginia.

Given under my hand this 24 th day of Oct., 19 60.

Address of celebrant 449 N Braddock Norman E Cooper (Person who performs ceremony sign here.)

V. S. 77

Figure 7. An example piece of data from the dataset Virginia Marriage Certificates October 1960.

Table I
Results for the deep learning models we tested

Model	mAP50	CRA	WRA	Use
YOLOv8	.995	N/A	N/A	Object Detection
EasyOCR	N/A	85.40%	58.99%	OCR on Machine Printed
TrOCR	N/A	17.16%	3.83%	OCR on Handwritten

V. Results

After 200 epochs which took less than 10 minutes our YOLO model achieved an mAP50 of .995; this number was first reached at 50 epochs however the training that followed helped to bring down the different loss types. Our model boasts a cls_loss of .2922, box_loss of .6015, and dfl_loss of .8358. The off the shelf EasyOCR model tested at an 85.4% Character Recognition Accuracy (CRA) and 58.99% Word Recognition Accuracy (WRA). The off the shelf TrOCR model tested at a 17.16% CRA and 3.83% WRA.

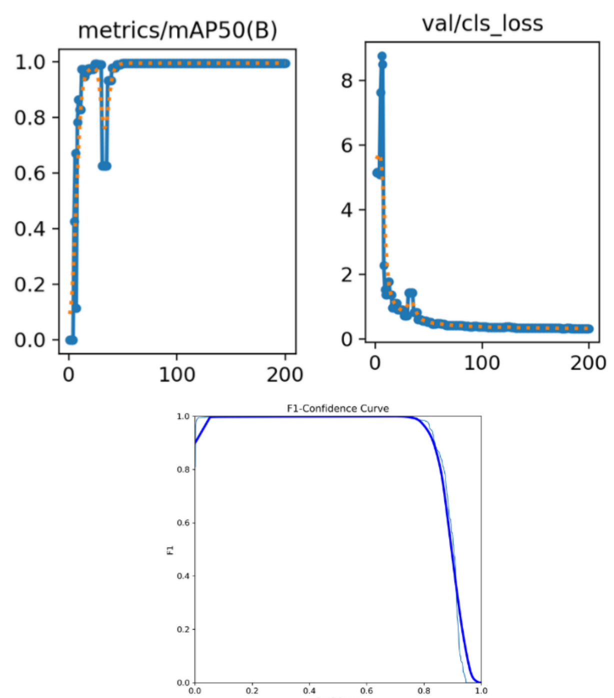


Figure 8. left mAP50 graph. right class loss graph. below F1 confidence curve

VI. Discussion

Our object detection network does a wonderful job as we expected thanks to the easy to recognize fields. This allows for quick modifications to pivot to recognizing a new document format. The labeling process takes around 2 hours for 20 pieces of data to train your object detection model. You can see in Fig. 8 how we reach .995 mAP50 at about 50 epochs and almost immediately before there is a big dip. This is because we implemented a dropout to reduce the likelihood of overfitting. After 50 epochs you can see the continued reduction of the cls_loss; the other loss metrics followed a similar trend.

Easy OCR performed relatively well off the shelf with the 85.4% CRA. You can see in fig. 5 an example of the data it is being expected to recognize and in fig. 6 is what the model predicted. For future research we suggest a small amount of transfer learning on your dataset to familiarize the model with the font/writing style common in the documents. On the opposite side of the spectrum the TrOCR model performed terribly. You can see in fig. 3 and fig. 9 examples of the types of images TrOCR is expected to recognize. This is much different from the typewriter text and can be challenging for experienced humans.

Dealing with cursive handwriting is difficult as there are various styles and levels of pen-man-ship or lack thereof. Low quality or hard to read records are inevitable so the field of study is forced to find a way around this which could be to toss the data or call in a person for the required human touch. One consideration is that this system is best used on document formats where data in ROI is either printed handwriting or typewritten. Further research will be required to understand the limitations of this system when it comes to handwritten text recognition.

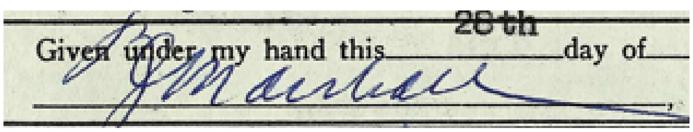


Figure 9. An example of why cursive writing can prove difficult to accurately recognize

With the increased power of deep learning data extraction systems we need to be mindful of how we use the data. In addition to accuracy metrics we must take into account privacy. Clear legal frameworks must be developed to help preserve the balance between legal protection and public accessibility[1]. Those deploying Handwritten Text Recognition technology on historical collections need to be critical of how they may follow or resist literary, historical, and cultural canons[4].

A logical next step in the research is to find a way to increase WRA using sophisticated post-processing systems to clean the data. The work completed by Sol et al.[5] should be the researched more; especially the post processing work done by their system. This increases the accuracy and complexity of the system which eventually becomes an efficient but expensive solution[3].

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